Improving Factual Reliability in Agent-Based Literature Surveys

Evaluated on Healthcare Literature using OpenManus

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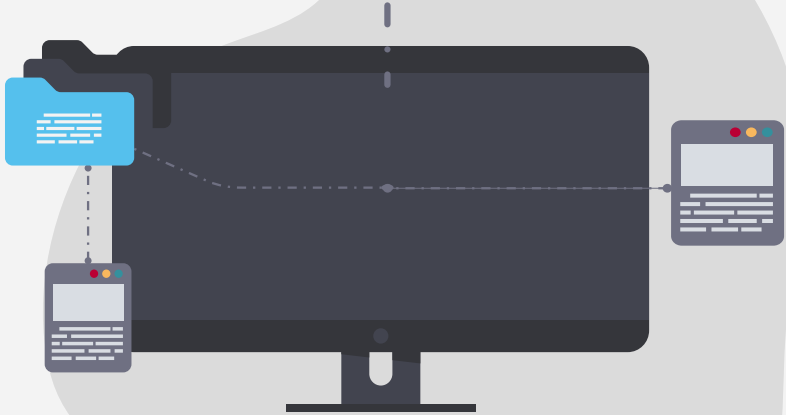

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Motivation

Trust and Reliability in AI Agents

- LLM-based agents are increasingly used for knowledge synthesis
- Applications span healthcare, finance, law, and academic research
- Agent outputs often sound confident but may be factually incorrect
- Lack of verification limits real-world deployment and trust

Problems in Literature Survey Agents

- Generate multi-step summaries across multiple papers
- Prone to:
    - Unsupported or hallucinated claims
    - Incorrect attribution to sources
    - Overgeneralization of limited evidence
    - Errors are subtle and difficult to detect automatically

Motivation

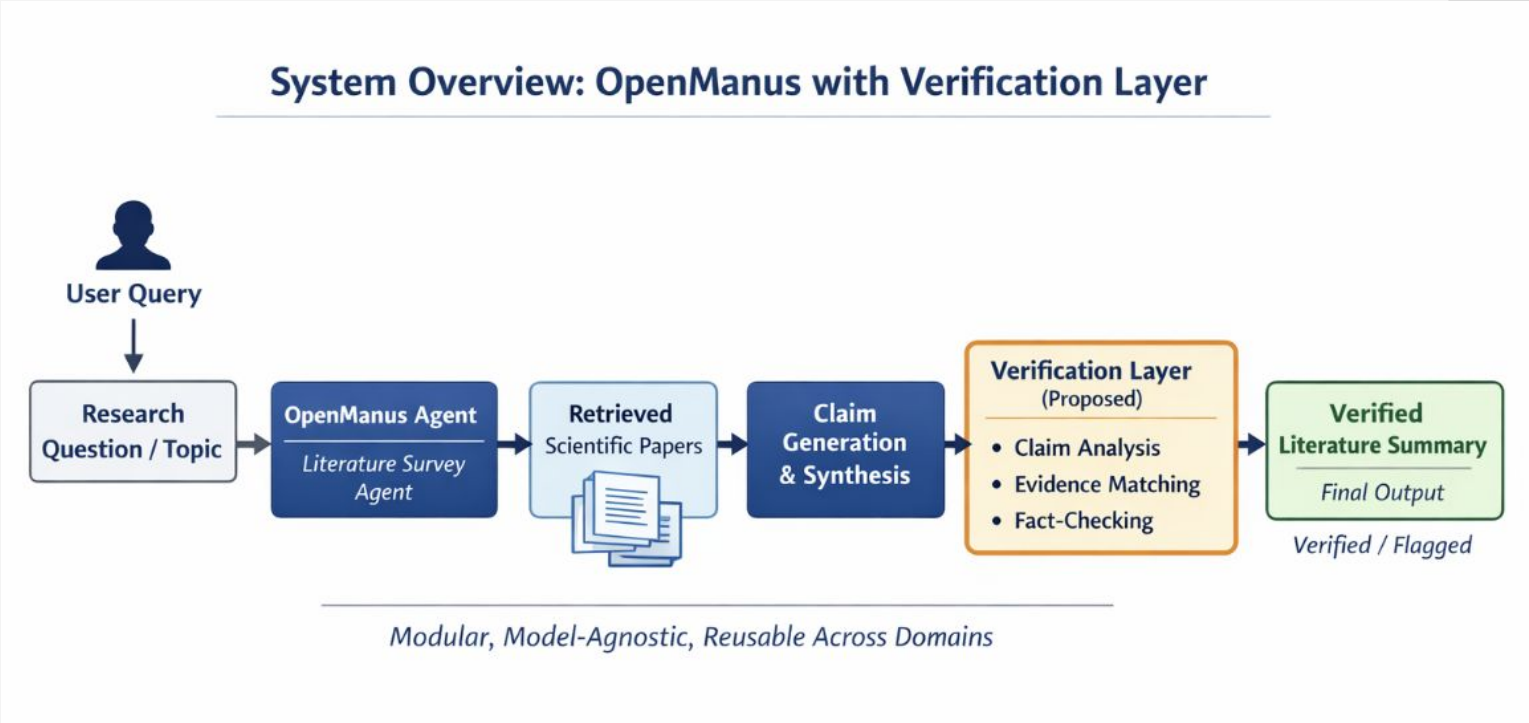
Why Focus on Healthcare Literature?

- Low tolerance for factual error and hallucinations
- Literature contains complex, nuanced claims:
  - Statistical results
  - Correlation vs causation
- Enables focused development by concentrating on one domain
- Serves as a rigorous stress test for verification methods
- If verification works on healthcare data, it is likely to generalize to other domains

Proposed Solution

- Analyze OpenManus architecture and agent pipeline
- Design and implement a verification layer for literature survey agents
- **Verification Layer:**
  - Added as a modular component in the OpenManus pipeline
  - Operates after agent reasoning and document retrieval
  - Evaluates:
    i. Generated claims
    ii. Retrieved evidence
    iii. Claim-evidence alignment
    iv. Flags unsupported or inconsistent claims before final output
- Integrate verification into the OpenManus workflow
- Apply and evaluate the system on healthcare literature
- Compare baseline agents with verification-enhanced agents

Proposed Solution



Output Example

Before: Baseline Literature Survey Agent

Generated Output

Drug X is associated with a significant reduction in blood pressure in patients with hypertension.

(Hallucination)

After: Verification-Enhanced Agent

Generated Output

Drug X is associated with a significant reduction in blood pressure in patients with hypertension.

- Claim Support Rate: **90%**
- Citation Accuracy: **80%**
- Evidence Match Quality: **75%**
- Hallucination Rate: **6%**

Evaluation Plan

- Compare baseline OpenManus agents with verification-enhanced agents
- Use identical queries and document sets for fair comparison
- Analyze factual correctness, citation quality, and claim support
- Few Important Metrics I plan on using:
  - ***Claim Support Rate*** - Percentage of generated claims that are supported by retrieved evidence
  - ***Citation Accuracy*** - Whether cited papers actually support the claims they are attached to
  - ***Evidence Match Quality*** - Strength of alignment between claims and retrieved passages
  - ***Hallucination Rate*** - Percentage of claims not grounded in any retrieved document
  - ***Human Evaluation*** - Manually reviewing the output using a rubric if needed

Related Background Research

- **Deep Research Agents: A Systematic Examination and Roadmap**
  (Huang et al., University of Liverpool, Oxford, UCL, Huawei Noah's Ark Lab)
  - Provides a comprehensive overview of research agents and their architectures
  - Highlights limitations in factual reliability and lack of verification mechanisms

- **LiRA: A Multi-Agent Framework for Reliable and Readable Literature Review Generation**
  (Go et al., University of Amsterdam, Elsevier, University of Copenhagen)
  - Introduces multi-agent collaboration to improve literature review quality
  - Inspires reliability-focused agent design but relies on agent coordination rather than explicit verification

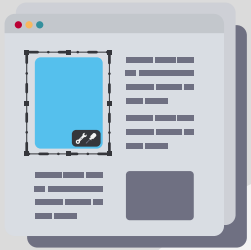
- **AIssistant: An Agentic Approach for Human-AI Collaborative Scientific Work**
  (Gaddipati et al., TIB, L3S Research Center)
  - Explores agent-based systems for scientific writing with human-in-the-loop verification
  - Motivates automated verification to reduce reliance on manual human oversight

Planned Project Milestones

- **Milestone 1: Understanding OpenManus & Literature Survey Agent**
  - Analyze OpenManus pipeline and components
  - Build initial healthcare literature survey agent
  - Identify gaps in output correctness and claim reliability

- **Milestone 2: Building the Verification Layer**
  - Design and implement modular verification system
  - Integrate with OpenManus agent workflow
  - Enable claim-evidence checking and flagging of unsupported claims

- **Milestone 3: Evaluation & Benchmarking**
  - Compare baseline vs verification-enhanced agent performance
  - Quantify improvements using claim support, citation accuracy, hallucination rate
  - Final refinements and system improvements based on results



Thank You!